

LONG TERM ANIMAL TRACKING SYSTEM

By

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Requirements

- Design an energy autonomous device that is capable of logging the position and activity (sleeping or awake) of a medium sized animal (e.g. sheep-size).
- The whole system needs to be as light as possible but not more than 1000 g.
- Use any kind or combination of harvesting devices you require but ensure a lifetime of 2 years or more with a datapoint at least every minute
- Use only commercially available components
- Logged data needs to be transferred to the local base station every few days. Alternatively you may use cellular or other available infrastructure.



Sensors



Activity Sensor

➤ Bosch Sensortec BMA280 (V = 2.4V)

Operating Mode	Current Consumption (μA)	Wake-up Time (ms)	Continuous Power (μW)
Normal Mode	130	0	312
Low Power Mode	1	1.3	2.4

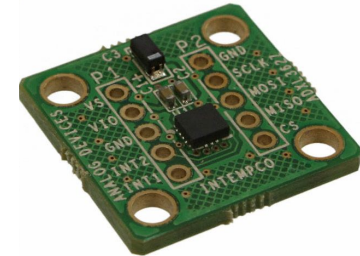
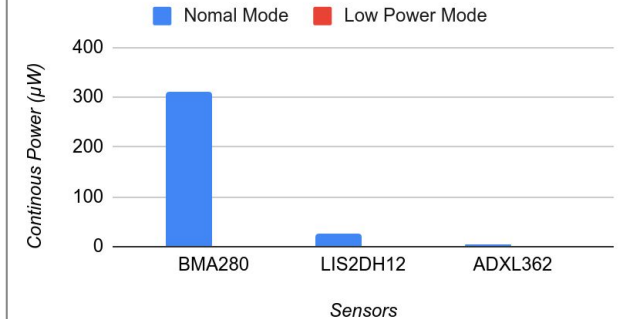
➤ STMicroelectronics LIS2DH12 (V = 2.5V)

Operating Mode	Current Consumption (μA)	Wake-up Time (ms)	Continuous Power (μW)
Normal Mode	11	1.6	27.5
Low Power Mode	0.5	0	1.25

➤ Analog Devices ADXL362 (V = 2V)

Operating Mode	Current Consumption (μA)	Wake-up Time (ms)	Continuous Power (μW)
Normal Mode	1.8	40	3.6
Low Power Mode	0.01	5	0.02

Power Consumption



Selected Module : Analog Devices ADXL362!

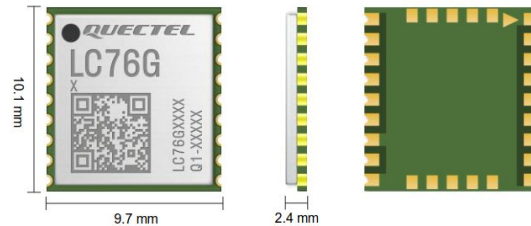
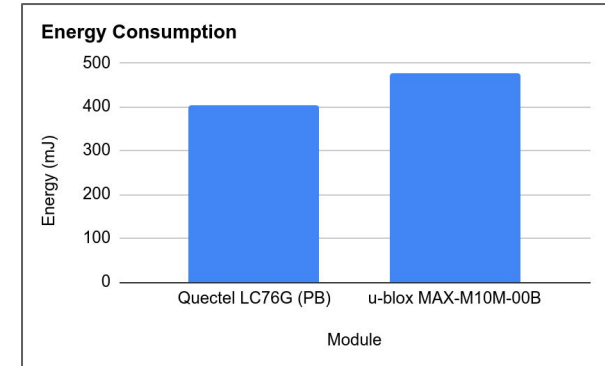
Global Positioning System

➤ Quectel LC76G (PB) (V = 1.8V)

Power Consumption (mW)	TTFF Cold Start (s)	Energy (mJ)	Processed Data per Fix (Bytes)
27	15	405	20-40

➤ u-blox MAX-M10M-00B (V = 1.8V)

Power Consumption (mW)	TTFF Cold Start (s)	Energy (mJ)	Processed Data per Fix (Bytes)
17	28	476	20-40



Selected Module : Quectel LC76G!

Data Transmission



LoRa Transceiver

➤ Semtech SX1262 (Baseline)

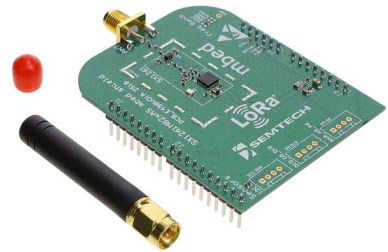
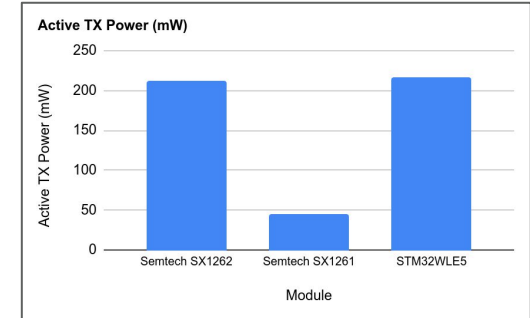
Vin Range (V)	Rx Current (mA)	RX Power (mW)	Wake-up Time cold (ms)	Active TX Power (mW)	Max Packet Size
1.8 - 3.7	4.6	8.28	3.5	212.4	255 Bytes

➤ Semtech SX1261 (Optimized)

Vin Range (V)	Rx Current (mA)	RX Power (mW)	Wake-up Time cold (ms)	Active TX Power (mW)	Max Packet Size
1.8 - 3.7	4.2	7.56	3.5	45	255 Bytes

➤ STM32WLE5 (SoC Module)

Vin Range (V)	Rx Current (mA)	RX Power (mW)	Wake-up Time cold (ms)	Active TX Power (mW)	Max Packet Size
1.8 - 3.6	5.4	9.72	150µs	216	255 Bytes



Selected Module : Semtech SX1261!

Power Budget



Worst Case Energy Requirement

Activity Sensor

Analog Devices ADXL362

Normal Mode:

$$t_{\text{active}} = 40\text{ms} = 0.04\text{s}$$

$$P_{\text{active}} = 3.6\mu\text{W} = 0.0036\text{mW}$$

$$E_{\text{active}} = 0.0036\text{mW} \times 0.04\text{s} = 0.000144\text{mJ}$$

Low Power Mode:

$$t_{\text{standby}} = 59\text{s}$$

$$P_{\text{standby}} = 0.02\mu\text{W} = 0.00002\text{mW}$$

$$E_{\text{standby}} = 0.00002\text{mW} \times 59\text{s} = 0.00118\text{mJ}$$

Position Sensor

Quectel LC76G

Normal Mode:

$$t_{\text{active}} = 15\text{s}$$

$$P_{\text{active}} = 33\text{mW}$$

$$E_{\text{active}} = 33\text{mW} \times 15\text{s} = 495\text{mJ}$$

Low Power Mode:

$$t_{\text{standby}} = 45\text{s}$$

$$P_{\text{standby}} = 43\mu\text{W} = 0.043\text{mW}$$

$$E_{\text{standby}} = 0.043\text{mW} \times 45\text{s} = 1.935\text{mJ}$$

Data Transmission

Semtech SX1261

Normal Mode:

$$t_{\text{active}} = 61.7\text{ms} = 0.0617\text{s}$$

$$P_{\text{active}} = 1.8\text{V} \times 24.0\text{mA} = 43.2\text{mW}$$

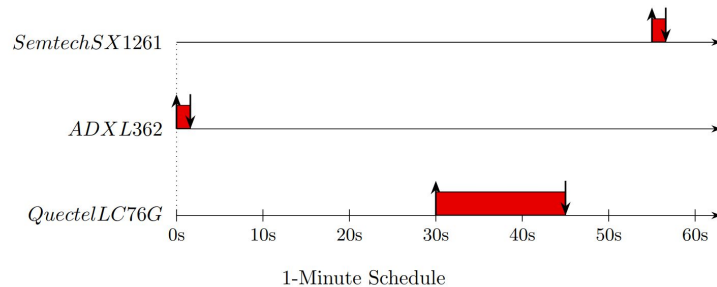
$$E_{\text{active}} = 43.2\text{mW} \times 0.0617\text{s} = 2.665\text{mJ}$$

Low Power Mode:

$$t_{\text{active}} = 59\text{s}$$

$$P_{\text{active}} = 1.8\text{V} \times 160\text{nA} = 0.288\mu\text{W} = 0.000288\text{mW}$$

$$E_{\text{active}} = 0.000288\text{mW} \times 59\text{s} = 2.682\text{mJ}$$



Worst case Energy: 495mJ

Energy Harvester



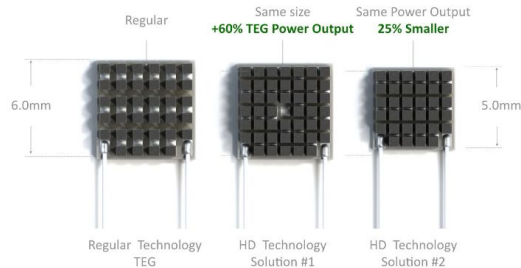
➤ TEC Microsystems 1MD06-097-05 TEG.

N = 97 High-Density Bismuth Telluride (BiTe) thermocouple pairs.

Size = 11mm x 11mm

Electrical Specifications	
Seebeck Coefficient per Couple	0.4mV/K
AC Resistance (ACR) per couple	0.0288 Ω (at 27°C)
Total Module Seebeck Coefficient (α_{TEG})	38.8 mV/K
Base Internal Resistance	2.79 Ω

HIGH DENSITY (HD) PELLETS PLACEMENT TECHNOLOGY EXAMPLE



➤ Performance Calculation: Single TEG ($\Delta T = 5K$)

$$\alpha_{TEG} = N \cdot 0.4 \text{ mV/K} = 97 \cdot 0.4 \text{ mV/K} = \mathbf{38.8 \text{ mV/K}}$$

$$V_{oc} = \alpha_{TEG} \cdot \Delta T = 38.8 \text{ mV/K} \cdot 5 \text{ K} = \mathbf{194 \text{ mV}}$$

$$R_{eff} \approx 1.15 \cdot (97 \cdot 0.0288 \Omega) \approx \mathbf{3.21 \Omega}$$

$$V_{load} = \frac{V_{oc}}{2} = \mathbf{97 \text{ mV}}$$

$$P_{max} = \frac{V_{oc}^2}{4 \cdot R_{eff}} = \frac{(0.194 \text{ V})^2}{4 \cdot 3.21 \Omega} \approx \mathbf{2.93 \text{ mW}}$$

➤ Performance Calculation: Five TEG Array (series)

$$V_{oc} \text{ (total)} = 5 \cdot V_{oc} \text{ (single)} = 5 \cdot 194 \text{ mV} = \mathbf{970 \text{ mV}}$$

$$R_{eff} \text{ (total)} = 5 \cdot 3.21 \Omega = \mathbf{16.05 \Omega}$$

$$V_{load} \text{ (total)} = \frac{V_{oc} \text{ (total)}}{2} = \mathbf{485 \text{ mV}}$$

$$P_{max} \text{ (total)} = \frac{V_{oc}^2 \text{ (total)}}{4 \cdot R_{eff} \text{ (total)}} = \frac{(0.970 \text{ V})^2}{4 \cdot 16.05 \Omega} \approx \mathbf{14.66 \text{ mW}}$$

Energy Storage



Storage

➤ NEXT104Z5.5V11.5X8.5F

Capacitance (F)	Max voltage (V)	Equivalent Series Resistance (ESR) (Ω)	Maximum Leakage Current (30 min) (mA)
0.1	5.5	< 16 at 1 kHz	0.14

➤ BZ015B603ZSB

Capacitance (F)	Max voltage (V)	Equivalent Series Resistance (ESR) (m Ω)	Maximum Leakage Current (30 min) (μ A)
0.06	5.5	80	10

➤ VOLTcraft LIR2032 (Button cell)

Capacity (mAh)	Nominal output voltage (V)	Lifetime	Battery type
45	3.6	< 400 cycles	Lithium rechargeable



Selected Module : LIR2032!

Time to charge

0.1F Supercapacitor

Supercapacitor Charging Time (0.1F, $V_{min} = 2V \rightarrow V_{max} = 4.5V$)

$$E = \frac{1}{2} C (V_{max}^2 - V_{min}^2)$$

$$E = \frac{1}{2} (0.1) (4.5^2 - 2^2) = 0.05 \times 16.25 = 0.8125 \text{ J}$$

$$t = \frac{E}{P_{out}}$$

$$t = \frac{0.8125 \text{ J}}{0.01319 \text{ W}} \approx 61.6 \text{ s}$$

0.06F Supercapacitor

Supercapacitor Charging Time (0.06F, $V_{min} = 2V \rightarrow V_{max} = 4.5V$)

$$E = \frac{1}{2} (0.06) (4.5^2 - 2^2) = 0.03 \times 16.25 = 487.5 \text{ mJ}$$

Button Cell

$$Q = 45 \text{ mA h} = 0.045 \times 3600 = 162 \text{ C}$$

$$E_{battery} = V \times Q = 3.6 \times 162 = 583.2 \text{ J}$$

Time to fully charge the cell from empty

$$t = \frac{E_{battery}}{P_{out}}$$

$$t = \frac{583.2 \text{ J}}{0.01319 \text{ W}} \approx 44216 \text{ s} \approx 12.28 \text{ h}$$

Time to charge after single burst:

$$I_{charge} = \frac{P_{out}}{V} = \frac{13.19 \text{ mW}}{3.6 \text{ V}} \approx 3.664 \text{ mA}$$

$$t_{recover} = \frac{\Delta Q_{used}}{I_{charge}} = \frac{0.1389 \text{ C}}{0.003664 \text{ A}} \approx 37.9 \text{ s}$$

Power Management



Power Management ICs

➤ e-peas DS-AEM30940

Minimum Input Voltage (Cold-Start) (mV)	Low-Voltage LDO Output (V)	High-Voltage LDO Output(V)	Efficiency (%)
380 @ 3 μ W	1.2 or 1.8 (20 mA)	1.8 to 4.1 (80 mA)	80-90

➤ Texas Instruments BQ25505

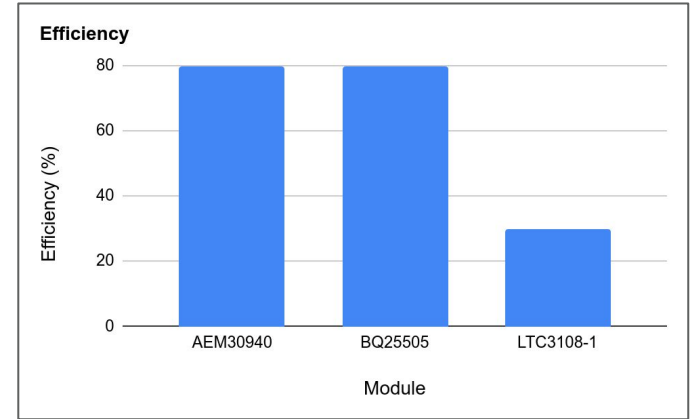
Minimum Input Voltage (Cold-Start) (mV)	Quiescent Current (nA)	Output Voltage to Battery (V)	Efficiency (%)
≥ 600	325	2 to 5.5	80-90

➤ Linear Technology LTC3108-1

Minimum Input Voltage (Cold-Start) (mV)	Selectable Main Output (VOUT) (V)	Integrated LDO Output (V)	Efficiency (%)
20	2.5, 3.0, 3.7, or 4.5	2.2 at 3mA	30

$$P_{out} = P_{max} \times \eta = 14.66 \text{ mW} \times 0.90 \approx 13.19 \text{ mW}$$

$$I_{out} = \frac{P_{out}}{V_{BATT}} = \frac{13.19 \text{ mW}}{4.5 \text{ V}} \approx 2.93 \text{ mA}$$



Selected Module : DS-AEM30940!

System



➤ STMicroelectronics STM32WLE5CCU6

48 MHz Arm Cortex-M4 | Integrated sub-GHz LoRa | UFQFPN48: 7 × 7 × 0.6 mm

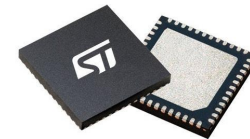
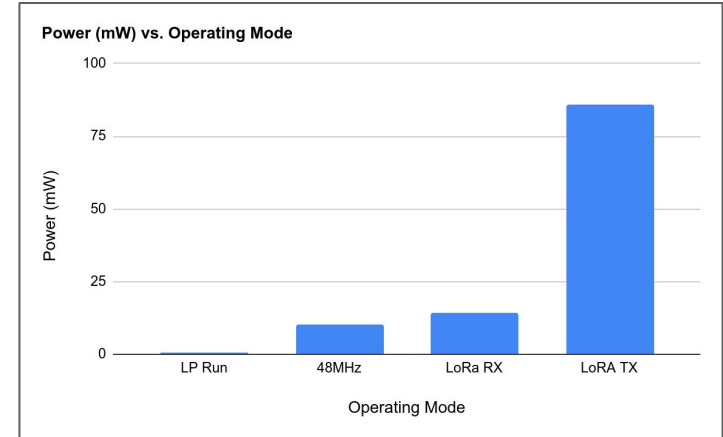
Operating Mode	Typical Current	Approx. Power
Stop 2 with RTC	1.07 μ A @ 3.0 V	3.21 μ W
Low-Power Run @ 2 MHz	220 μ A @ 3.0 V	0.66 mW
Run @ 48 MHz (SMPS)	3.45 mA @ 3.0 V	10.35 mW
LoRa Receive	4.82 mA @ 3.0 V	14.46 mW
LoRa TX @ +14 dBm	26 mA @ 3.3 V	85.8 mW

MCU energy per 1-minute cycle

≈ 0.71 mJ/min

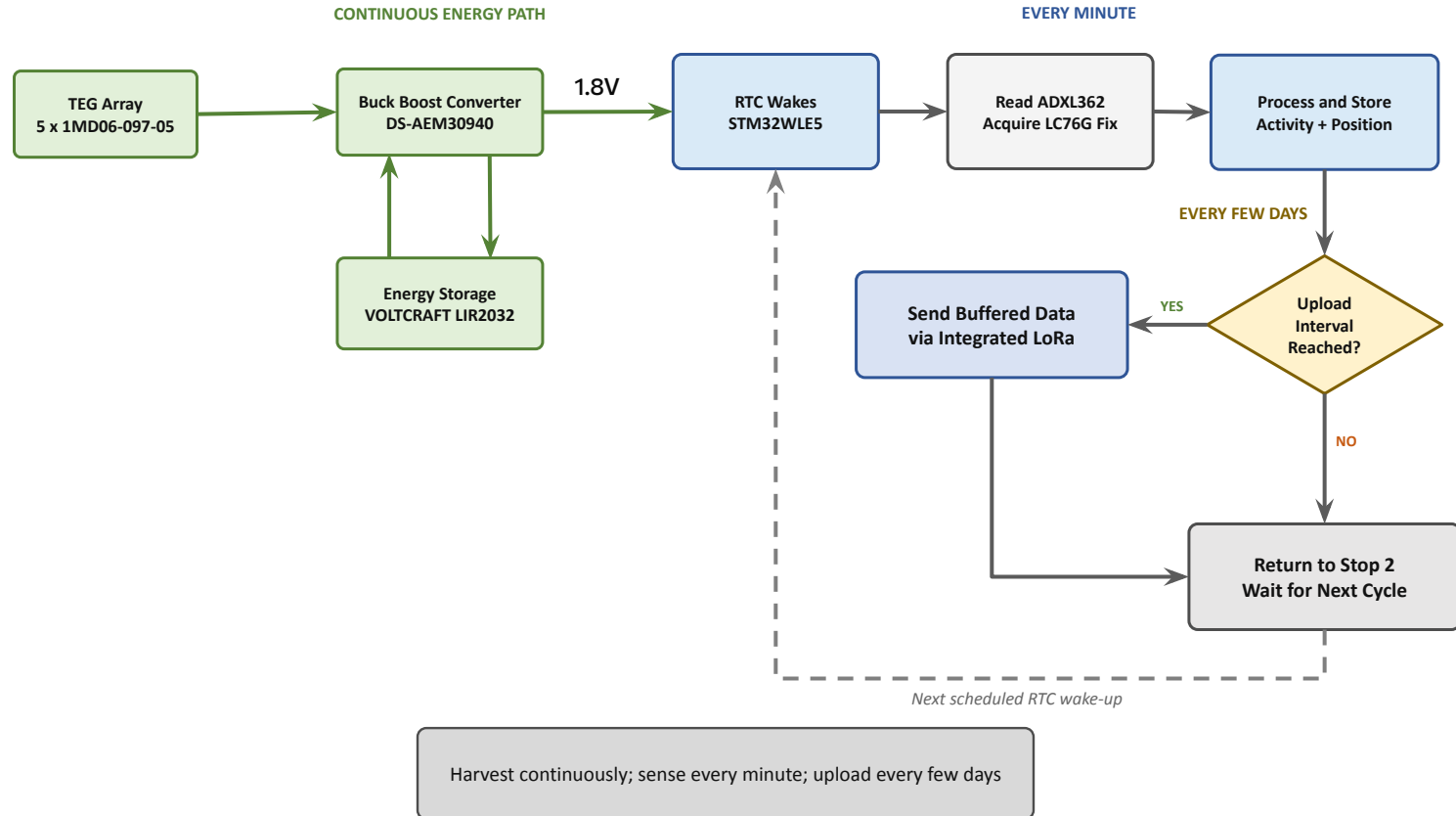
Assumption: 50 ms at 48 MHz; remaining 59.95 s in Stop 2 with RTC

Selected SoC: STM32WLE5CCU6 — MCU



UFQFPN48 package

System Operation Flowchart

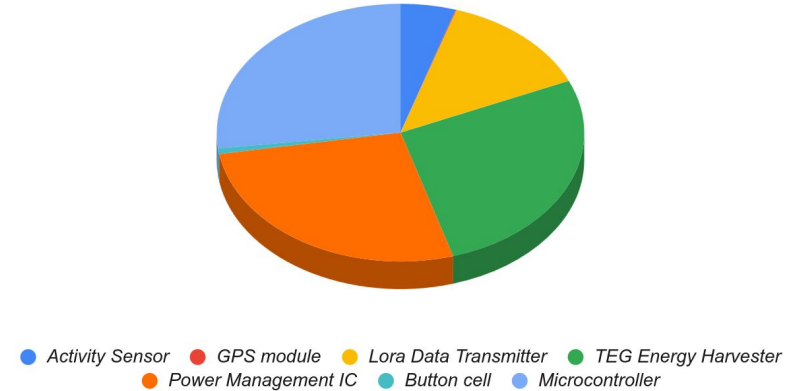


Bill of Material



S No	Part	Description	Cost (€)	Weight (mg)
1	ADXL362	Activity Sensor	6.45	18
2	Quectel LC76G	GPS module	6.31	0.3
3	Semtech SX1261	Lora Data Transmitter	8.3	50
4	1MD06-097-05	TEG Energy Harvester	65	100
5	DS-AEM30940	Power Management IC	70	100
6	LIR2032	Button cell	1.99	2.6
7	STM32WLE5CCU6	Microcontroller	2.97	100
Total			161.02	370.9

Weight (mg)



References

- <https://www.semtech.com/products/wireless-rf/lora-connect/sx1261>
- https://e-peas.com/product/aem30940-v/?gad_source=1&gad_campaignid=23171289009&gbraid=0AAAAADiWiEqOsqDu9IMosPBnGTH7h0BCY&gclid=CjwKCAjwsfzSBhB5EiwAOGyqSXf3LhlpvFodPFor5oAllq2KH4orMLWE4cCj-Qt1YU6Co6J3GmNhihoC9xIQAvD_BwE
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- <https://www.analog.com/media/en/technical-documentation/data-sheets/ADXL362.pdf>
- <https://www.quectel.com/product/gnss-lc76g-series/>
- <https://www.u-blox.com/en/product/max-m10-series>
- <https://www.st.com/resource/en/datasheet/stm32wle5cc.pdf>
- <https://www.tec-microsystems.com/product/1md06-097-05/>

Thank You